

Islanded green energy system optimal analysis using PV, wind, biomass, and battery resources with various economic criteria

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ABSTRACT

The main goal of this paper is to design an efficient renewable energy system that meets the required electricity demands. Consequently, it is essential to find the most cost-effective hybrid system that can reduce energy costs and provide access to the required electricity generation. An international school in New Administrative Capital in New Cairo, Egypt was chosen as a study area for the proposed system. Solar, wind, and biomass resources are abundant at the chosen location throughout the year. Using the HOMER (hybrid optimization model for electric renewables) software, eight distinct models of renewable energy hybrid systems were designed, simulated, and optimized to meet the required load in this study. For wind and solar resources, the National Aeronautics and Space Administration (NASA) provided the input data; for biomass resources, real-time field data were used for the selected study site. In this study, lithium-ion and lead acid batteries were used to choose the most cost-effective option. The hybrid power system's PV, wind, and biomass generators were utilized to meet the load demand. The overabundance of energy requests was utilized to charge the battery banks as required when accessible to cover the load requirements during times of low energy production.

1. Introduction

Energy crises, rising electricity prices, and lack of grid access are the most pressing challenges in emerging Asia, the Middle East, and Sub-Saharan Africa. The greatest solution to tackle developing countries' economic and energy difficulties is to construct hybrid renewable energy systems in off-grid or grid-connected modes. The development of hybrid renewable energy systems, on the other hand, involves significant technical and economic obstacles [1].

The issue of the random nature of renewable energy supplies is typically addressed by using a variety of energy sources, which is exacerbated by the fact that these resources are entirely dependent on unpredictable weather conditions, which do not always produce energy with the precision that is needed. Energy from wind and solar sources may be utilized in conjunction with each other to create a more predictable and stable energy mix than if they were used separately [2–6]. A higher percentage of renewable energy may now be used, opening the door for the development of interconnected renewable energy systems. This is made feasible by creating renewable energy micro-grids and

incorporating Energy Storage Systems (ESSs) with renewable energy systems [7–14].

A few decades ago, along with other techniques, lead-acid batteries were the most often utilized energy storage devices for electric power system purposes. As opposed to that, lithium-ion batteries have lately acquired popularity due to their reduced maintenance needs, enhanced safety, increased gravimetric energy densities, and distinctive volumetric features. These batteries are smaller and less weight than lead-acid batteries [15–19].

Recently, lithium-ion batteries have been widely used for energy storage in many applications, such as hybrid power microgrids, electric vehicles, and medical devices, due to features such as high energy density, high efficiency, superior capacity, and long-life cycle when compared to other types of dry batteries. However, performance deterioration and safety concerns occur because of the cell's anomalous temperature increase. As a result, lithium-ion batteries must be kept in a certain temperature range throughout operation to ensure safety and improve performance. To minimize the maximum temperature of the battery, an effective battery thermal management system (BTMS) is

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required. As a result, Hamed et al. [20] presented an updated comprehensive overview of research on external thermal management methods to provide a thorough grasp of the most recent essential technologies employed for external BTMS. Furthermore, practical instructions for expanding the applicability of such technologies have been proposed.

Building an integrated photovoltaic (BIPV) system is gaining popularity among academics due to its environmental friendliness and ability to save land resources. Combining storage batteries and BIPV can improve overall system adaptability, which is attractive for distributed renewable energy applications. How to optimally dispatch the hourly energy flow of PV panels, storage batteries, and power networks depending on a building load, on the other hand, is critical and understudied. As a result, Xinxin et al. [21] developed a multi-restricted condition nonlinear optimization model for a BIPV-battery storage hybrid system under diverse building loads on a clear day. The optimization model was solved using the `fmincon` function in MATLAB code. The total minimal daily cost of the hybrid system, electric price, and carbon price was considered as objective functions in the optimization to obtain the optimal operation strategy of hourly power distributions of PV, battery, and grid for daily building consumption.

Kerboua et al. [22] sought to create an ideal scaling simulation model for an off-grid photovoltaic-wind hybrid power system at an Algerian industrial location. Their hybrid system was sized using the loss of power supply probability technique. The case study's technical and economic evaluation revealed that the storage system accounted for

most of the total investment cost of the hybrid system. The investment cost analysis revealed the single best configuration for each size of the battery bank. The hybrid system's combination resulted in an energy cost that was quite competitive with most European countries.

The large demand for energy in Egypt due to the large increase in population and the need to wide using renewable power sources as a substitute source for traditional fossil fuel energy so, the encouragement to use the off-grid system to feed the large projects and new cities is considered an important matter to reduce large demand of energy on the electricity grid. So, renewable power hybrid systems in off-grid places are considered a great method to feed different loads. Due to the expensive cost of hybrid power systems, their use has been restricted in many developing nations. So, the scope of the present work is concerned with finding the best model that can be used to offer the required load at the lowest cost. We built a variety of hybrid energy system models for this article. The most cost-effective choice is chosen by comparing lithium-ion and lead-acid batteries with the different hybrid energy systems.

2. Methodology and description of the proposed hybrid power systems

The case study for the suggested system in the current research was an international school in New Administrative Capital in New Cairo, Egypt. The chosen location has good year-round access to biomass,

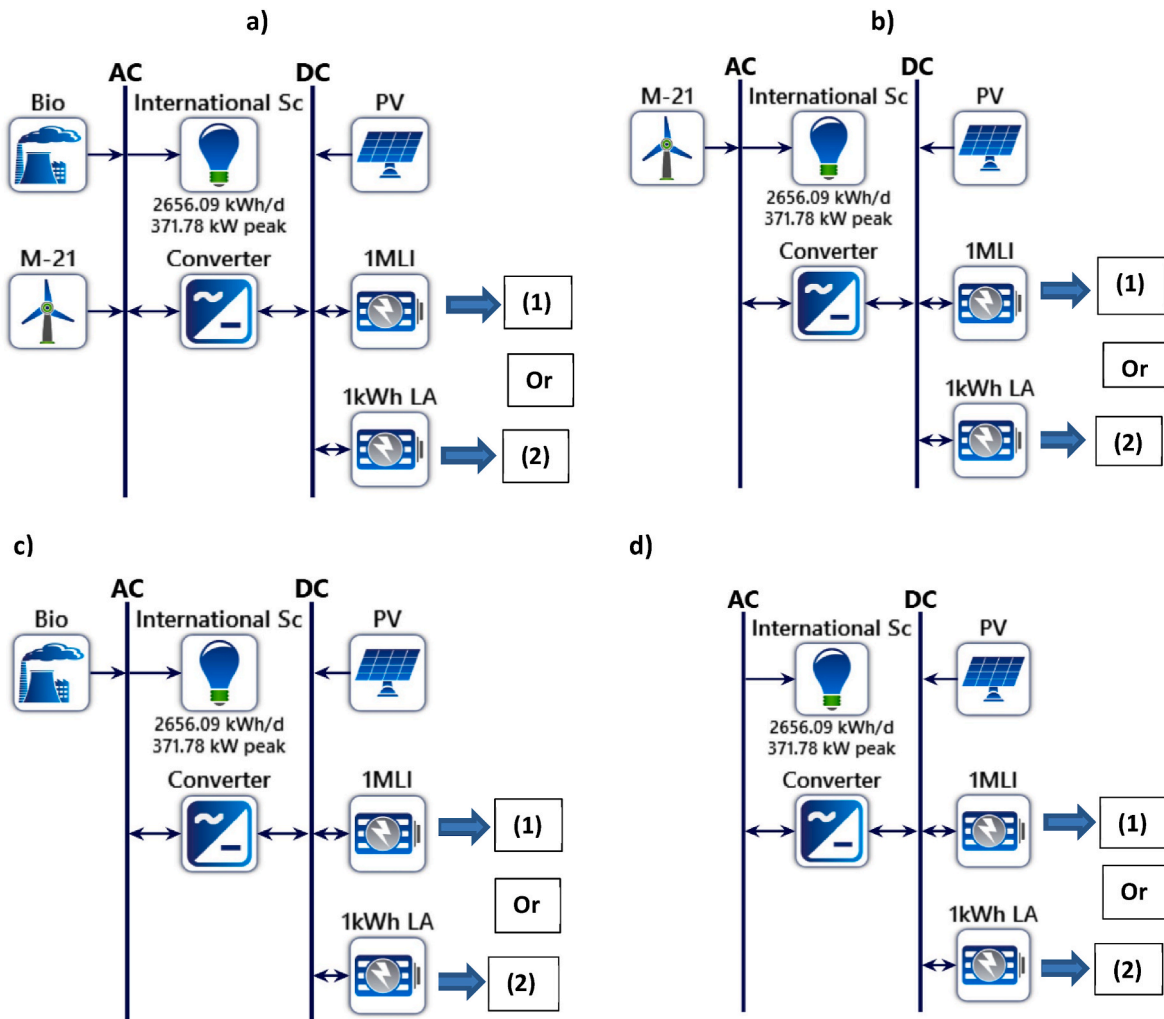


Fig. 1. A variety of hybrid systems powered by renewable energy that use (1) Lithium-ion batteries or (2) Lead Acid batteries: a) PV-biomass-wind-batteries hybrid system configuration; b) PV-batteries-wind hybrid system configuration; c) PV-biomass-batteries hybrid system configuration; d) PV-batteries system configuration.

wind, and solar resources. A biomass generator, a wind turbine, a battery bank, a converter, and an electrical load with a mean value of 2656 kWh/day make up the majority of the suggested energy system.

The objective of this article is to build a price-effective sustainable power system that can supply the necessary power demand for an international school. In order to select the most economical hybrid system for them, this paper presents eight renewable energy power systems. The first four hybrid power systems are PV-biomass-wind-batteries, PV-batteries-wind, PV-biomass-batteries, and PV-batteries with a type of batteries of a lithium-ion with 1 MWh of power reserve and 600V of nominal voltage as presented in Fig. 1-(1). The second four hybrid power systems are PV-biomass-wind-batteries, PV-batteries-wind, PV-biomass-batteries, PV-batteries but with a type of batteries of lead acid with 51.64 kWh of energy storage and 12V of nominal voltage as shown in Fig. 1-(2).

The technical and economic studies were conducted using the HOMER software program, which supports the best design of renewable hybrid systems. The program simulates the operation of the system during its entire lifetime, the necessary input data correspond to initial expenditures and other running costs [23].

2.1. Load profile

The research area's mean load demand per day is utilized to simulate the yearly load profile per hour. The area under research has an average load of roughly 2656 kWh per day, a maximum load of 371.78 kW, and a load factor of 0.3. Fig. 2 displays the selected site's monthly average daily load. From that figure, it's observed that the peak load is in June and the minimum load is in December. For e.g., in June, the load slightly decreased from 56.16 kW at 0 o'clock until reached 55.87 kW at 2 o'clock, then it slightly increased from 55.87 kW at 2 o'clock until reached 56.5 kW at 4 o'clock, then it increased from 56.5 kW at 4 o'clock until reached to the maximum value at 15 o'clock, and finally, it decreased until reached 58.15 Kw at the end of the day. Fig. 3 illustrates the selected site's annual average load over the hour and day of the year. The load's highest values are depicted in that figure in the final days of June. The load's lowest values are depicted in that figure in the final days of December. For the rest days of the year, the load varies between the minimum and maximum values.

2.2. Renewable energy systems

2.2.1. Solar generation system

Irradiance from the sun and temperature are utilized in this study to determine how much power solar PV panels can produce. Equation (1) [24] is used to compute the output power of PV panels.

$$P_{PV} = C_{PV} DF_{PV} \left(\frac{R_T}{R_{T,STC}} \right) [1 + \alpha_P (T_p - T_{p,STC})] \quad (1)$$

Where:

C_{PV} : The rated output power capacity in kW of PV arrays.

DF_{PV} : Derating factor of PV (%).

R_T : The amount of solar radiation that incident the PV array (in kilowatts per m²).

$R_{T,STC}$: The fall radiation at standard test conditions (1 kW/m²).

α_P : Power temperature coefficient in percent/degree Celsius.

T_p : The temperature of the PV cell in the current time step (in degrees Celsius).

$T_{p,STC}$: The PV cell temperature at 25 °C under typical test conditions.

Equation (2) [24] is used to calculate the temperature of a PV cell:

$$T_p = \frac{T_a + (T_{p,NOT} - T_{a,NOT}) \left(\frac{R_T}{R_{T,NOT}} \right) \left[1 - \frac{\eta_p \eta_{p,STC}}{\tau_s \alpha_s} \right]}{1 + (T_{p,NOT} - T_{a,NOT}) \left(\frac{R_T}{R_{T,NOT}} \right) \left(\frac{\alpha_P \eta_{p,STC}}{\tau_s \alpha_s} \right)} \quad (2)$$

Where:

T_a : The ambient temperature in (degrees Celsius).

$T_{p,NOT}$: The nominal operating temperature (NOT) of the PV cell in (degrees Celsius).

$T_{a,NOT}$: The ambient temperature that is used to determine the nominal temperature (20 °C).

$R_{T,NOT}$: The solar radiation at which the nominal temperature is defined (0.8 kW/m²).

$\eta_{p,STC}$: The peak power efficiency under standard test conditions (%).

τ_s : The solar transmittance of any PV array-covering material (%).

α_s : The sun-based absorptance of the PV array (%).

The NASA Surface Meteorology and Solar Energy website [25] provided the solar resource for the international school in the New Administrative Capital of New Cairo, Egypt, at 29°59.8' N latitude and

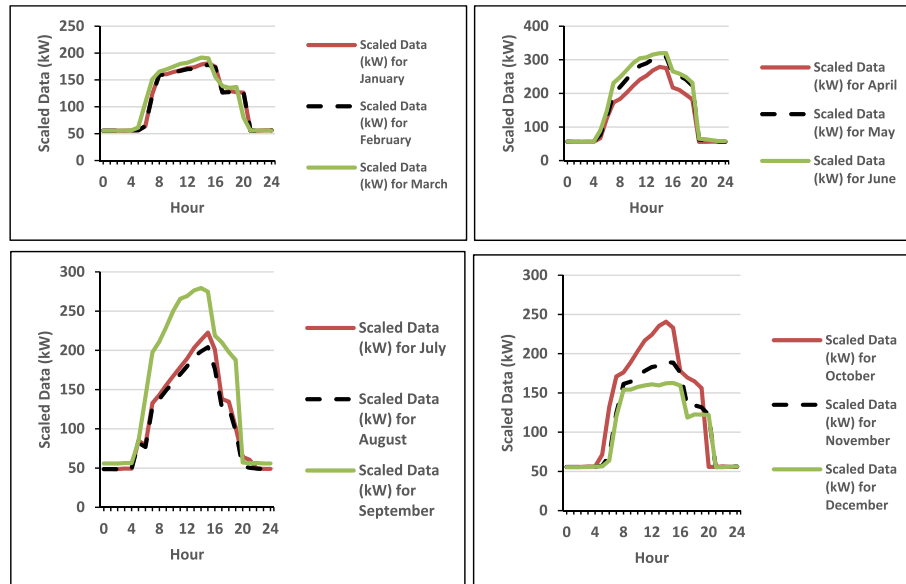


Fig. 2. The selected site's monthly average daily load.

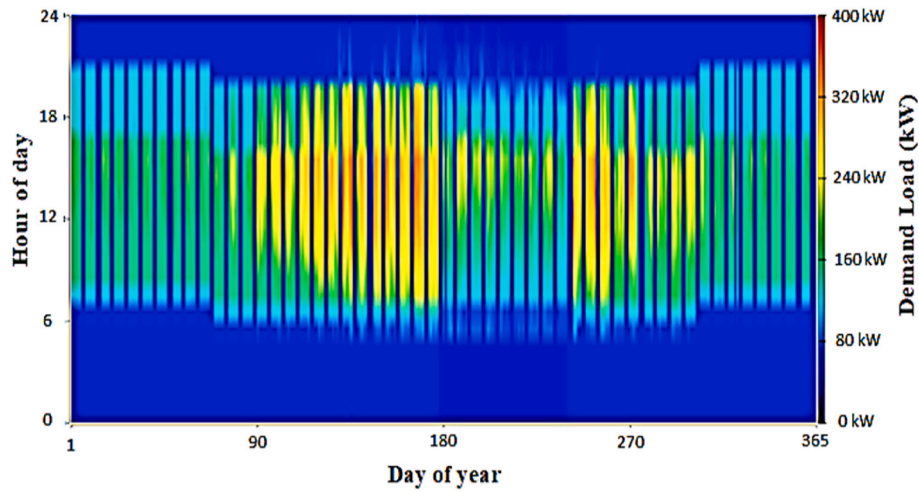


Fig. 3. The selected site's annual average load over the hour and day of the year.

31°26.2' E longitude.

Fig. 4 depicts the scaled data of global solar radiation at the selected site. According to this graph, the maximum value of solar radiation for the study area is in June (7.69 kWh/m²/day), while the minimum value is in December (3 kWh/m²/day). The selected site's annual average solar radiation is 5.35 kWh/m²/Day. This site receives a significant amount of solar radiation, indicating that the solar energy system is an important power source.

2.2.2. Wind generation system

HOMER uses a three-step process to calculate the power output of the wind turbine in each time step. First, HOMER computes the wind velocity at the wind turbine's hub height. The wind turbine's power output is then calculated at that wind speed and air density. Finally, HOMER modifies the power output value to account for the actual air density.

HOMER calculates the wind velocity at the wind turbine's hub height at each time step using the inputs you specify on the Wind Resource page and the Wind Shear entry. If you use the logarithmic law, the hub height wind velocity is calculated using equation (3) [26]:

$$V_h = V_a \cdot \frac{\ln\left(\frac{Z_h}{Z_s}\right)}{\ln\left(\frac{Z_a}{Z_s}\right)} \quad (3)$$

Where:

V_h : Wind velocity at the wind turbine's hub height in meters per second.

V_a : The wind velocity at anemometer height in meters per second

Z_h : The wind turbine's hub height (m).

Z_a : Height of the anemometer (m).

Z_s : Long of surface roughness (m).

If the power law is used, HOMER computes the wind speed at the wind turbine's hub height using equation (4) [26]:

$$V_h = V_a \cdot \left[\frac{Z_h}{Z_a}\right]^\gamma \quad (4)$$

Where:

γ : The power law exponent.

After calculating the hub height wind velocity, the power curve of the wind turbine is used to determine the estimated output power of the wind turbine at that wind velocity under standard temperature and pressure conditions.

Power curves usually describe wind turbine effectiveness under standard temperature and pressure conditions (STP). To adapt to real circumstances, multiply the power value expected by the power curve by the air density ratio, as shown in equation (5) [26]:

$$P_w = \left(\frac{\rho_a}{\rho_n}\right) \cdot P_n \quad (5)$$

where:

P_w : The output power of a wind generator (kW).

P_n : The power output of a wind generator at normal pressure and temperature (kW).

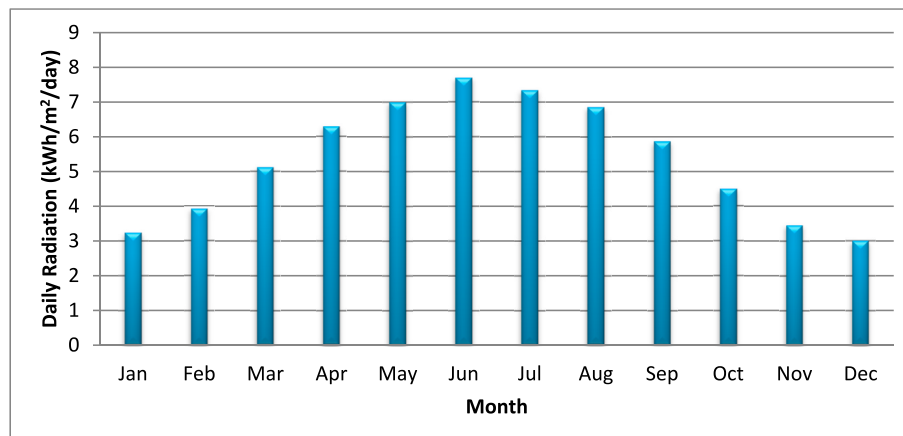


Fig. 4. Monthly average solar global radiation of the selected site.

ρ_a : The density of air (kg/m³).

ρ_n : Density of air at normal pressure and temperature (1.225 kg/m³).

In this work, the wind generator is designed with a hub height of 31.8 m, a rated power of 100 kW, and a rotor radius of 10.5 m. The mean monthly wind velocity information is gathered from the NASA resource website based on the location of the selected area [25]. The selected site's yearly average wind velocity is 5.56 m/s. The monthly average wind velocity data of the selected site is illustrated in Fig. 5. According to that figure, the maximum wind speed is observed in April, while the minimum wind velocity is observed in August. The power characteristics of the selected location's wind turbines are shown in Fig. 6. From that figure, the predicted output power of the wind turbine can be found.

2.2.3. Generation of biomass power system

A biochemical operation (fermentation or anaerobic digestion) or a thermochemical operation (gasification, pyrolysis, or direct combustion) can be used to generate electricity from biomass. These operations produce a gaseous or liquid fuel [27]. This study is based on the utilization of residues from agricultural crops in Egypt. The monthly average available biomass data of the selected site is shown in Fig. 7. The mean data of biomass source in the selected site is approximately (1.15ton/-day). The maximum available biomass supply is observed in August and September as shown in Fig. 7. In the present research, a biogas Genset (auto size) with a rated capacity of 500 kW is used. The capital cost and replacement cost per unit kW of the biomass system are assumed to be 3000\$ and 1250\$ respectively. The operating and maintenance costs per unit kW are estimated at 0.10 \$/op.hr. The operating lifetime of the engine is 20,000 h. The gasifier engine converts solid biomass into gas with a high degree of efficiency. Engines with internal combustion (IC) are employed in biomass power systems. The method used to determine how much fuel the IC engines utilize is based on an article [27]. Fig. 8 shows the fuel consumption versus generator output power of the selected site. The relation between fuel consumption (kg/hr) and generator output power (kW) is linear. The generator can't run at very low loads since it operates at a minimum load ratio of 50% of its rated power. Equation (6) is used to get the generator output power (P_b) [28].

$$P_b(t) = \left(\frac{N_g}{F} \right) \left[\frac{\eta_g \times H_w \times B(t)}{H_g} - F_n P_g \right] \quad (6)$$

Where:

P_b: Output power of Biomass generator (kW).

F_n: Fuel consumption at No-load (kg/h per kW rated).

F: Small amounts of fuel consumption (kg/h per kW output).

H_g: Lower producer gas heating values (MJ/kg).

H_w: Lower wood heating values (MJ/kg).

B: Biomass supply in 1 h (kg/h).

N_g: Number of generators.

η_g : Gas efficiency.

P_g: Rated power of generator (kW).

For a 500 kW generator, F₀ is assumed to be 0.1 (kg/hr/kW) and F₁ to be 2 (kg/h/kW). Equation (7) [28] can be used to obtain the hourly consumption rate (m_w(t)).

$$m_w(t) = \frac{1}{\eta_g} \left(\frac{H_g}{H_w} \right) (N_g F_n P_g + F P_b(t)) \quad (7)$$

Table 1 shows the gas emission analysis of the biomass power system. There are very small amounts of carbon dioxide, carbon monoxide, and nitrogen oxides. There aren't any unburned hydrocarbons, particulate matter, or sulfur dioxide.

3. Power inverter model

To enable feeding the AC load when required, a power inverter is used to convert the DC power generated by the PV panels and the DC batteries into AC power. The maximum load demand determines the size of the power inverter to be used.

Equation (8) can be used to determine the input power of the inverter (P_v):

$$P_v(t) = \frac{P_L(t)}{\eta_v} \quad (8)$$

Where η_v and P_L(t) are inverter efficiency and load power respectively.

An inverter's efficiency indicates how much DC power is converted to AC electricity. Some power is lost as heat, and some stand-by power is consumed to keep the inverter operational.

The efficiency of the inverter unit (η_v) might be expressed as in equation (9) [29]:

$$\eta_v = \frac{P_o/P_i}{\frac{P_o}{P_i} + \left[1 - 99 \left(\frac{10}{\eta_a} - \frac{1}{\eta_b} - 9 \right)^2 \right] + \frac{P_o/P_i}{\eta_b}} \quad (9)$$

Where η_a and η_b : The inverter efficiencies at 10 and 100% of nominal power.

P_o: Output power of the inverter.

P_i: Input power the of inverter.

In this paper, the inverter used has efficiency 95%.

4. The batteries bank system modeling

In any hybrid energy system, batteries are considered an essential component. The batteries can store excess energy from renewable en-

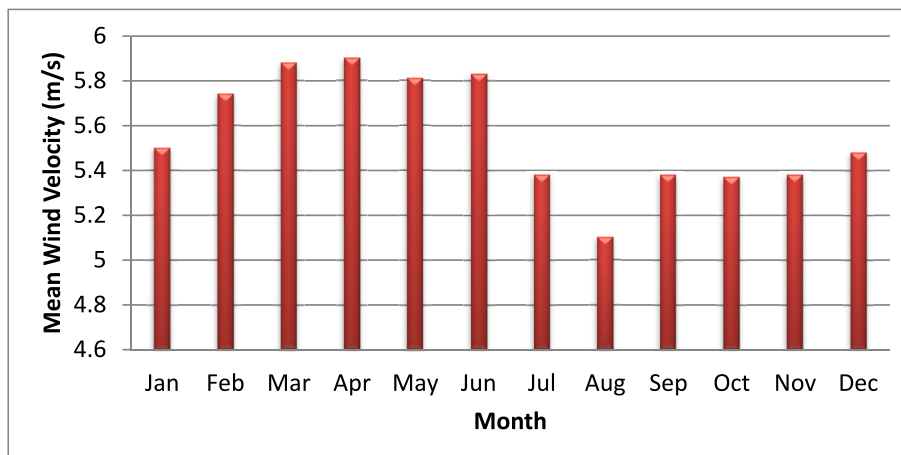


Fig. 5. Monthly mean wind velocity data of the selected site.

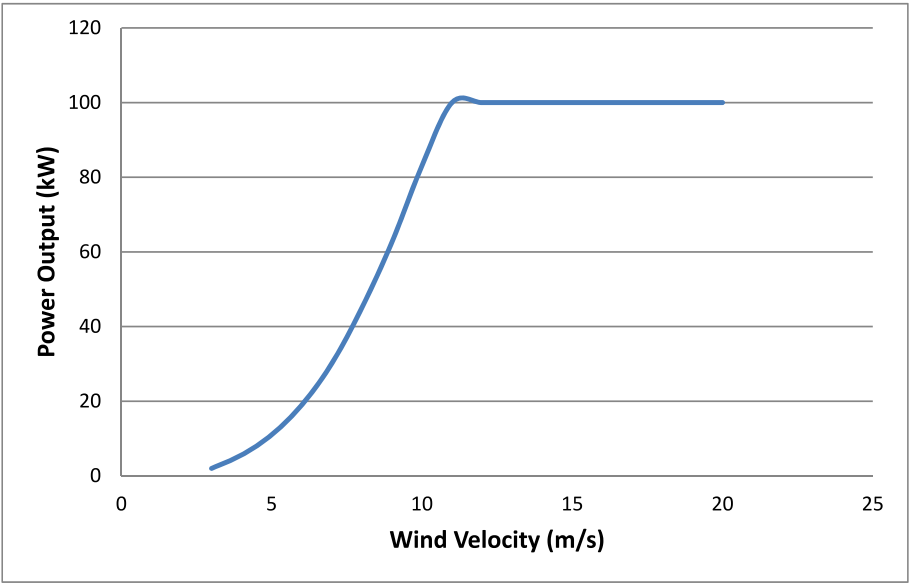


Fig. 6. Power characteristics of the selected location’s wind turbines.

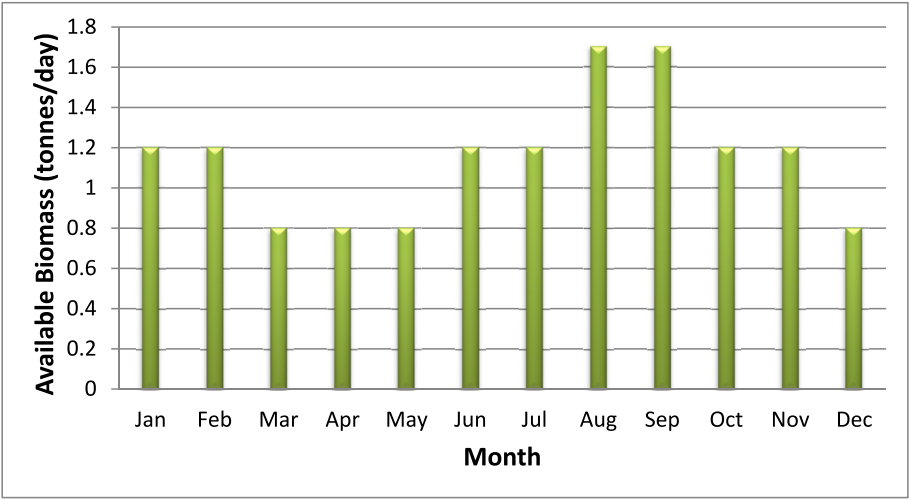


Fig. 7. The selected location’s monthly average available biomass data.

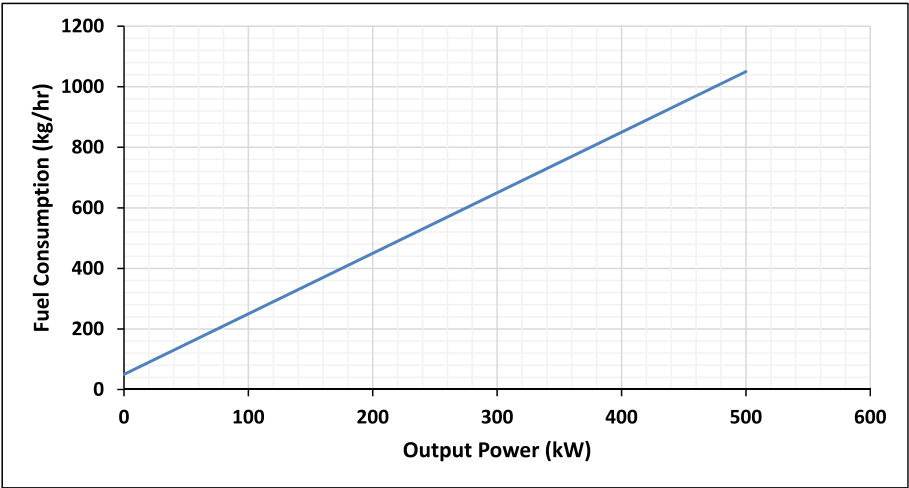


Fig. 8. Comparing fuel consumption to the selected site’s generator output power.

Table 1
Gas emission analysis of the biomass power system.

Pollutant	Quantity	Unit
Carbon Dioxide	-0.679	kg/yr
Carbon Monoxide	0.432	kg/yr
Unburned Hydrocarbons	0	kg/yr
Particulate Matter	0	kg/yr
Sulfur Dioxide	0	kg/yr
Nitrogen Oxides	0.270	kg/yr

ergy sources to feed the load when it is needed. Equation (10) [30] can be used to determine the batteries number (N_B) needed for this study:

$$N_B = \frac{E_d * n_d}{V_B * AH * DOD} \quad (10)$$

Where n_d : The autonomy days.

E_d : The daily demand for energy.

V_B : The battery's electric potential.

AH : The ampere-hour of battery.

DOD : The depth of discharge in the battery system.

In this paper, we used two types of batteries with various hybrid energy systems.

The first type of battery used in this research is a lithium-ion battery and the second type is a lead acid battery. The hybrid system's battery specifications are presented in Table 2.

This research also shows a comparison between the lithium-ion battery and lead acid battery when charged with various hybrid energy systems. This comparison helped us to determine the optimum type of battery and the minimum number of batteries that can be used to achieve the least net present cost (NPC) of the system.

5. Economic and optimization model

In this part, the NPC and the Levelized cost of energy (LCOE) concepts are computed throughout this research for the different configurations of renewable energy hybrid systems with types of batteries of lithium-ion and also with lead acid batteries to determine the most economic model. The LCOE is computed by using equation (11) [28,31]:

$$LCOE = \frac{CRF * NPC}{E_{Load}} \quad (11)$$

Where E_{Load} is the total load served in kWh/year and CRF is the capital recovery factor.

The capital recovery factor (CRF) is computed by equation (12) [32]:

$$CRF = \frac{r(1+r)^Y}{(1+r)^Y - 1} \quad (12)$$

Where r is the interest rate per year and Y is the project lifetime in years.

The interest rate per year (r) is obtained by Equation (13) [33]:

$$r = \frac{r' - f}{1 + f} \quad (13)$$

Where r' is the nominal interest rate in percentage and f is the inflation rate per year in percentage.

In the present research, the lifetime of the project is set to be 25 years, the interest rate is 8.3% according to the central bank of Egypt (CBE) [34] and the inflation rate is 4.829% according to CBE [35].

The following equation (14) is used to determine the NPC [36]:

$$NPC = N_S C_S + N_W C_W + N_{gen} C_B + N_{bat} C_{bat} \quad (14)$$

Where C_S , C_W , C_B , and C_{bat} are the price of the solar system (per unit), the wind turbine system (per unit), the biomass plant (per unit), and the battery bank (per unit) respectively. N_S , N_W , N_{gen} , and N_{bat} are the number of PV modules, wind turbines, generators, and battery banks respectively.

The cost of each component can be computed by using equation (15) [31]:

$$C = C_c + C_R + C_{O\&M} + C_F - S \quad (15)$$

Where, C_c is the capital price of a system component (\$) which includes the system design costs, engineering, installation, and balance-of-system (BOS). The price to replace a system component in dollars is called C_R . The salary of operators, insurance, maintenance, and inspections are included in the $C_{O\&M}$ cost (\$). The biomass power system's fuel cost per year in dollars is called C_F . S is the salvage value, which refers to the value that will remain in a power system component at the end of the project's lifespan.

C_F is obtained by using equation (16) [31]:

$$C_F = C_B * M_t \quad (16)$$

Where C_B is the cost of the biomass fuel and M_t is the biomass generator's total wood consumption in kilograms per year, as determined by equation (17) [31]:

$$M_t = \sum_{t=1}^T m_w(t), T = 8760 \quad (17)$$

Where T is the operating time (hr).

The salvage value (S) is determined by the replacement cost (C_R) instead of the capital cost (C_c).

S is computed by using equation (18) [26]:

$$S = C_R \left(\frac{R_r}{R_c} \right) \quad (18)$$

Where R_r and R_c represent the component's remaining life at the end of the project's lifespan (in years) and the component's lifespan (in years), respectively.

R_r is computed using equation (19) [26]:

$$R_r = R_c - (R_p - R_{rep}) \quad (19)$$

Where R_{rep} and R_p are the Replacement cost duration (in years) and project lifespan (in years), respectively.

R_{rep} is calculated using equation (20) [26]:

$$R_{rep} = R_c * INT \left(\frac{R_p}{R_c} \right) \quad (20)$$

INT: A function that returns the integer portion of a real number.

6. Results and discussions

Table 3 and Table 4 show optimization results of the different types of hybrid energy systems with lithium-ion batteries and lead acid

Table 2
The hybrid system's batteries' general specifications.

	lithium-ion battery	Lead acid battery
Nominal electric potential (V)	600V	12V
Nominal energy (kWh)	1000 kWh	1 kWh
Nominal capacity (Ah)	1670Ah	83.4Ah
Roundtrip efficiency (%)	90%	80%
Maximum charging current (A)	1670A	16.7A
Maximum discharging current (A)	5000A	24.3A
The first phase of charge (%)	100%	100%
Minimal level of charge (%)	20%	40%
Lifetime (years)	15 years	10 years
Throughput (kWh)	3000000 kWh	800 kWh

Table 3

Optimization results of the various types of hybrid energy systems with lithium-ion batteries.

Hybrid Energy Systems	PV (kW)	Wind Turbine (kW)	Biomass (kW)	No. of Lithium-Ion Batteries (1 MWh)	Converter (kW)	NPC (\$)	LCOE (\$)	Operating Cost (\$/yr)	Initial Capital Cost (\$)
PV/Wind/Biomass/Batteries	626	100	500	2	384	6.23 M	0.382	93,174	4.66 M
PV/Wind/Batteries	1014	100	–	5	419	8.95 M	0.549	160,927	6.24 M
PV/Biomass/Batteries	756	–	500	5	398	9.79 M	0.600	165,186	7.01 M
PV/Batteries	1514	–	–	6	436	11.4 M	0.699	195,044	8.11 M

Table 4

Results of optimization of the different types of hybrid energy systems utilizing lead acid batteries.

Hybrid Energy Systems	PV (kW)	Wind Turbine (kW)	Biomass (kW)	No. of lead acid batteries (1 kWh)	Converter (kW)	NPC (\$)	LCOE (\$)	Operating cost (\$/yr)	Initial capital Cost (\$)
PV/Wind/Biomass/Batteries	698	100	500	2784	392	6.70 M	0.411	143,906	4.28 M
PV/Wind/Batteries	1306	100	–	6016	512	8.46 M	0.519	187,475	5.30 M
PV/Biomass/Batteries	970	–	500	6244	446	9.16 M	0.562	191,963	5.93 M
PV/Batteries	1326	–	–	9531	424	11.1 M	0.681	285,743	6.30 M

batteries respectively. The rated capacity (kW) of each component and the number of batteries for each type of hybrid energy system are mentioned in these tables. The net present cost (NPC), the Levelized cost of energy (LCOE), the operating cost, and the initial capital cost of each hybrid energy system is illustrated in these tables. From the data in these tables, the least number of batteries used, the least NPC, the least LCOE, the least operating cost, and the least initial capital cost are observed for the hybrid system of PV/Wind/Biomass/Batteries. The NPC and LCOE of hybrid energy systems (PV/Wind/Biomass) with lithium-ion batteries are less than the same hybrid energy system but with lead-acid batteries. So, in this research, the optimum hybrid system is PV/Wind/Biomass with lithium-ion batteries.

The NPC and the LCOE of the optimum energy system are 6.23 million \$ and 0.382 \$ respectively. From these tables, it can be noticed that the total number of batteries used in the PV/Wind/Biomass hybrid energy system with lithium-ion batteries is two batteries only, but the total number of batteries used in the PV/Wind/Biomass energy system with lead acid batteries is 2784 batteries. So, the least number of batteries is observed for the optimum hybrid energy system (PV/Wind/Biomass) with lithium-ion batteries.

Fig. 9 shows the NPC of the various hybrid energy systems. The lowest NPC is observed at (PV/Wind/Biomass) hybrid systems with

lithium-ion batteries. The PV/batteries system has the largest NPC.

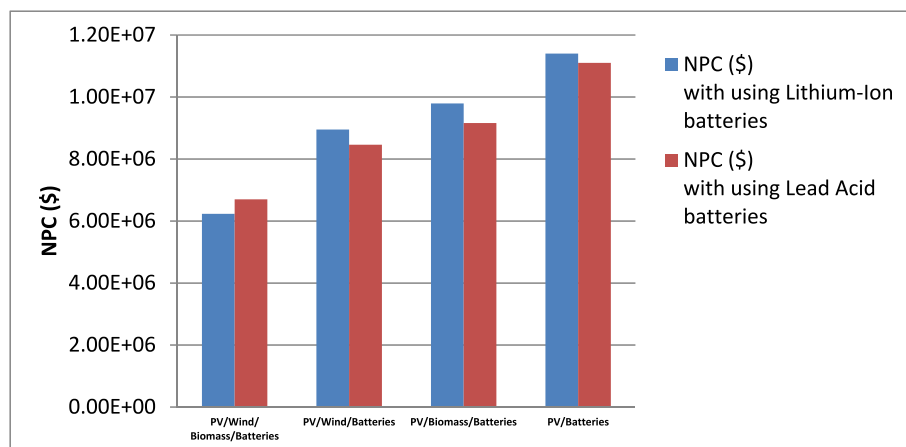
Fig. 10 shows the LCOE of the different hybrid energy systems. The (PV/Wind/Biomass) hybrid system with lithium-ion batteries has the least value of LCOE. The PV/batteries system has the largest LCOE.

Table 5 shows the Cost summary of the optimum PV-wind-biomass hybrid system with lithium-ion batteries. That table provides specifics regarding the capital cost, cost of replacement, operating cost, salvage cost, and resource cost of each part of the optimum hybrid energy system. The total costs of each part of the optimum hybrid system and also the total NPC of the system are displayed in that table.

Fig. 11 shows each component's costs in detail in the ideal hybrid system. From this figure, the maximum capital cost is observed at the flat plate PV component. The maximum operating cost, replacement cost, and salvage cost are observed for 1 MWh lithium-ion batteries. The only component which has the resource cost is the biogas Genset (biomass). The maximum total cost is observed at 1 MWh lithium-ion batteries.

Table 6 shows the electrical summary of input and output data of each part of the ideal PV-wind-biomass hybrid power system with lithium-ion batteries and lead acid batteries.

Table 7 shows the production summary of each part of the ideal PV-wind-biomass hybrid power system. The most amount of energy that was produced by the solar photovoltaic system in this study was 74.4%

**Fig. 9.** NPC of the various hybrid energy systems.

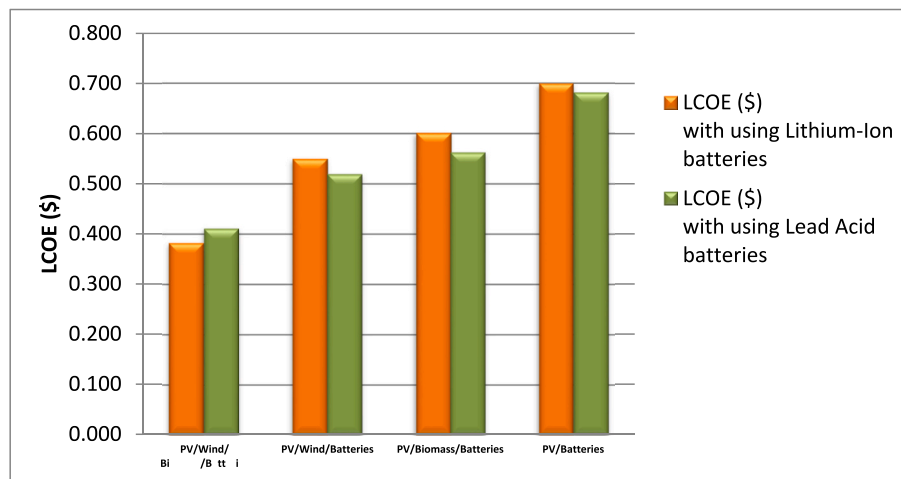


Fig. 10. LCOE of the various hybrid power systems.

Table 5

Cost summary of the optimum PV-wind-biomass hybrid energy system with lithium-ion batteries.

Name	Capital	Operating	Replacement	Salvage	Resource	Total
1 MWh Li-Ion batteries	\$1.40 M	\$336,492	\$858,861	-\$206,696	\$0.00	\$2.39 M
Biogas Genset (Biomass)	\$1.50 M	\$231,338	\$0.00	-\$181,666	\$363,613	\$1.91 M
Flat plate PV	\$1.56 M	\$105,293	\$0.00	\$0.00	\$0.00	\$1.67 M
System Converter	\$115,178	\$0.00	\$70,658	-\$17,005	\$0.00	\$168,831
Wind Turbine XANT M – 21 [100 kW]	\$80,000	\$6730	\$0.00	\$0.00	\$0.00	\$86,730
System	\$4.66 M	\$679,853	\$929,519	-\$405,367	\$363,613	\$6.23 M

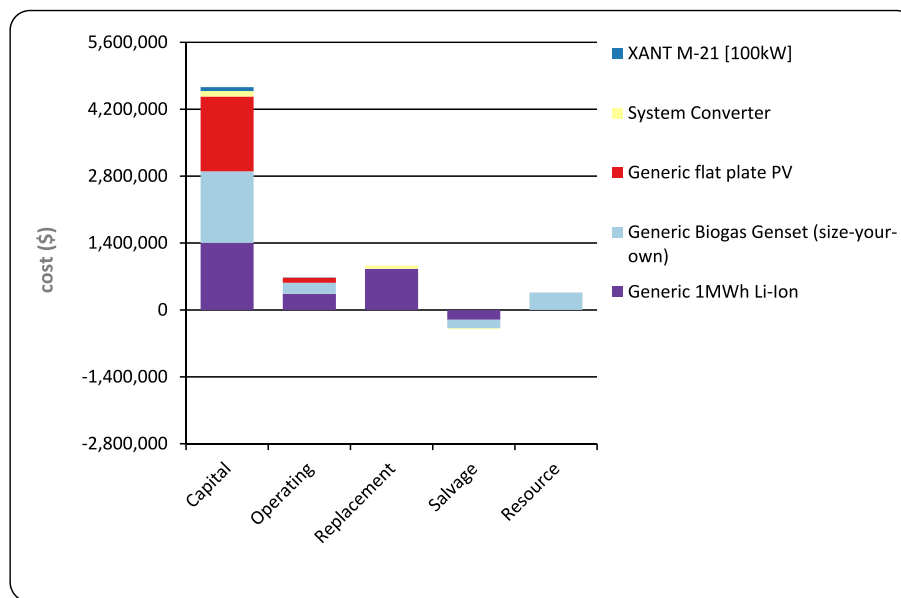


Fig. 11. Costs of each part in the ideal hybrid power system.

of the total amount of electricity produced. The minimum energy is produced from biomass by percent 4.84% of total electricity production. The energy produced by Wind turbines is by percent 20.8% of total electricity production. The excess electricity than load is estimated by a 26.8% increase in total production.

Fig. 12 displays the electric generation per month of the ideal hybrid energy system. From this figure, the maximum electricity production is observed in June but the minimum energy production is noticed in December.

Fig. 13 displays the annual power generation from biomass over 24 h

of the day of the optimum hybrid power system. From this figure, the minimum output electrical power and the maximum value are observed at 250 kW and 267 kW respectively. The annual electrical production of Biogas Genset is 68,767 kWh/yr. The fuel consumption per year and the specific fuel consumption are 216 tons/year and 2.20 kg/kWh respectively. The mean electrical efficiency is 29.8%. The Number of Starts and the Operational Life of Biogas Genset are 250 starts/yr and 72.7 year respectively. The total hours of operation are 275 h per year.

Fig. 14 shows the annual output power from the photovoltaic system over 24 h daily of the optimum hybrid power system. From this figure,

Table 6

Electrical summary of input and output data of each part of the ideal PV-wind-biomass hybrid power system with lithium-ion batteries and lead acid batteries.

Input and output data	Value	Input and output data	Value
Biogas Genset (Biomass) rowhead			
Electrical Production	68,767 kWh/yr	Specific Fuel Consumption	2.20 kg/kWh
Average Power production	250 kW	Fuel Energy Input	231,128 kWh/yr
The least Power production	250 kW	Average power Efficiency	29.8%
The most Power production	267 kW	Operation hours	275 h/yr
Annual consumption of fuel	216 tons/yr	Count of Starts	250 starts/yr
Daily consumption of fuel	0.592 tons/day	lifetime of operation	72.7 yr
Hourly consumption of fuel	0.0247 tons/hour	Capacity Factor	1.57%
Carbon Dioxide emission	−0.679 kg/yr	initial production Cost	88.4 \$/hr
Emission of Carbon Monoxide	0.432 kg/yr	Marginal Production Cost	0.286 \$/kWh
Nitrogen Oxides emission	0.270 kg/yr		
Flat plate PV rowhead			
Minimum production	0 kW	Rated Power	626 kW
Maximum production	626 kW	Average Power Output	121 kW
Penetration of PV	109%	Average Energy Output per day	2893 kWh/d
Operating Hours	4385 h/yr	Capacity Factor	19.3%
LCOE	0.0940 \$/kWh	Total Energy Generation per year	1,055,918 kWh/yr
Wind Turbine (XANT M-21 [100 kW]) rowhead			
The least Power production	0 kW	Total Rated Power	100 kW
The most Power production	100 kW	Average Power Output	33.6 kW
Wind Penetration	30.4%	Capacity Factor	33.6%
Operating Hours	7404 h/yr	Total Energy Generation per year	294,731 kWh/yr
LCOE	0.0175 \$/kWh		
1 MWh Li-Ion batteries rowhead			
Batteries	2.00 qty.	Annual Losses	31,635 kWh/yr
Size of String	1.00 batteries	Annual Production	300,217 kWh/yr
Strings in Parallel	2.00 strings	Autonomy	14.5 h
Bus Voltage	600 V	Cost of storage wear	0.246 \$/kWh
Mean Energy Cost	0 \$/kWh	Nominal Energy	2000 kWh
Energy Input	316,240 kWh/yr	Useable Energy	1600 kWh
Energy Output	284,811 kWh/yr	Lifetime Production	4,503,261 kWh
Storage Scarcity	206 kWh/yr	Expected Lifetime	15.0 yr
1 kWh Lead Acid batteries rowhead			
Batteries	2784 qty.	Annual Losses	68,980 kWh/yr
Size of String	1.00 batteries	Annual Production	308,662 kWh/yr
Strings in Parallel	2784 strings	Autonomy	15.1 h
Bus Voltage	12 V	Cost of storage wear	0.419 \$/kWh
Mean Energy Cost	0 \$/kWh	Nominal Energy	2786 kWh
Energy Input	344,731 kWh/yr	Useable Energy	1672 kWh
Energy Output	276,075 kWh/yr	Lifetime Production	2,227,200 kWh
Storage Scarcity	325 kWh/yr	Expected Lifetime	7.22 yr
System Converter rowhead			
Operation time	7489 h/yr	Average Power Output	76.6 kW
Annual Energy Output	671,235 kWh/yr	The least Power Output	0 kW

Table 6 (continued)

Input and output data	Value	Input and output data	Value
Annual Energy Input	706,563 kWh/yr	The most Power Output	341 kW
Annual Energy Losses	35,328 kWh/yr	Capacity Factor	20.0%
Power	384 kW		

Table 7

Production summary of each part of the ideal PV-wind-biomass hybrid power system.

Component	Annual Energy Generation (kWh/yr)	Percent
Flat plate photovoltaic	1,055,918	74.4%
Biogas Genset (Biomass)	68,767	4.84%
Wind Turbine (XANT M – 21 [100 kW])	294,731	20.8%
Total	1,419,416	100%
Excess Electricity	380,759	26.8%

the maximum and average output power is noticed at 626 kW and 121 kW respectively. The mean output energy and the total energy production of the photovoltaic system is 2893 kWh/day and 1,055,918 kWh/yr respectively. The total hours of operation are 4385 h per year.

Fig. 15 shows the annual power output from wind turbines over 24 h of the day of the optimum hybrid power system. The percentage of wind penetration is 30.4%. The maximum and mean output power in this figure are observed at 100 kW and 33.6 kW respectively. The total energy production of wind turbines is 294,731 kWh/yr. The total hours of operation are 7404 h per year.

Fig. 16 displays the annual state of charge (in percent) of lithium-ion batteries for the optimal hybrid energy system over a 24-h period. The batteries' initial and least states of charge are noted to be 100% and 20%, respectively. The input and output energy of batteries is 316,240 kWh/yr and 284,811 kWh/yr respectively. The energy loss is about 31,429 kWh/yr. The annual throughput is 300,217 kWh/yr.

7. Conclusion

To identify the best hybrid energy system for the chosen site, this work employs a variety of models of hybrid energy systems with a variety of components, including solar, wind, biomass, and battery systems. In this research, two different types of batteries are used. The two types of batteries are lithium-ion batteries and lead-acid batteries. The key results of this study can be summarized as follows: a hybrid power system (PV/Wind/Biomass) with lithium-ion batteries has a lower net present cost (NPC) and levelized cost of energy (LCOE) than a similar hybrid energy system with lead acid batteries. A PV/Wind/Biomass hybrid power system with lithium-ion batteries is ideal for the selected location. The optimum hybrid energy system has the least NPC and the least LCOE with values of 6.23 million \$ and 0.382 \$ respectively. The best hybrid power system (PV/Wind/Biomass with lithium-ion batteries) is shown to have the lowest operating cost among the eight models evaluated in this study. The PV/Wind/Biomass hybrid energy system only uses two lithium-ion batteries, whereas the same suggested hybrid power system for feeding the study's chosen site uses 2784 lead acid batteries. Therefore, the optimal hybrid energy system (PV/Wind/Biomass) using lithium-ion batteries is observed to have the fewest number of batteries. The excess electricity than load in this study is estimated by a 26.8% increase from total production.

Credit author statement

AMRO AMEEN: substantial contribution to conception and design; ELSAYED TAG-ELDIN: substantial contribution to acquisition of data;

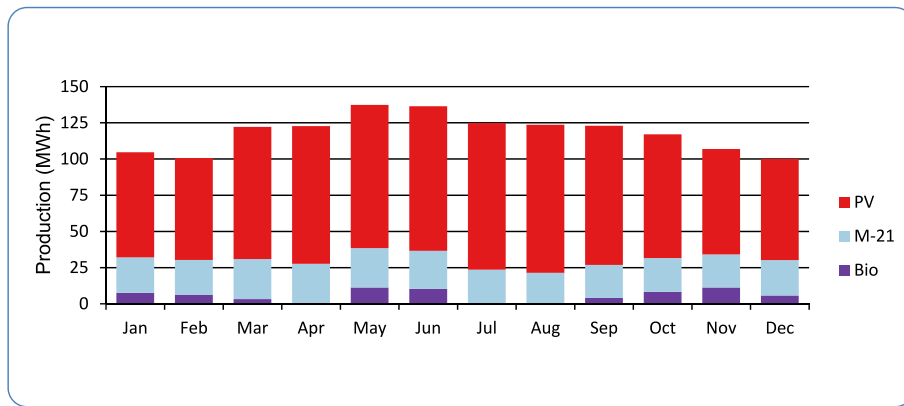


Fig. 12. Monthly electric production of the ideal hybrid energy system.

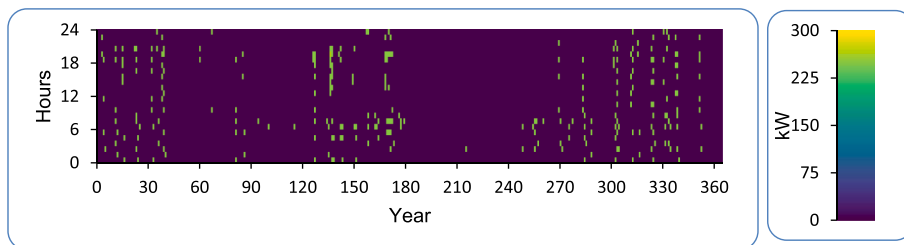


Fig. 13. Annual power generation from biomass over 24 h of the day of the optimum hybrid energy system.

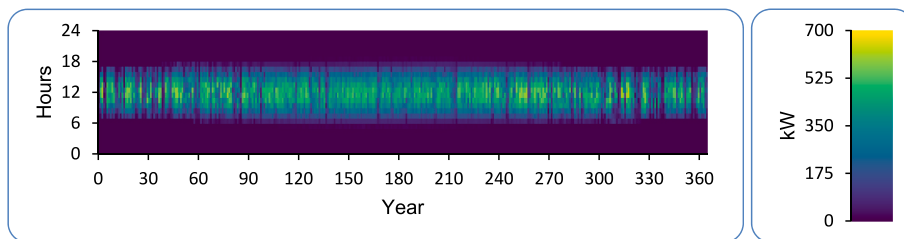


Fig. 14. Annual power output from PV system over 24 h of the day of the optimum hybrid energy system.

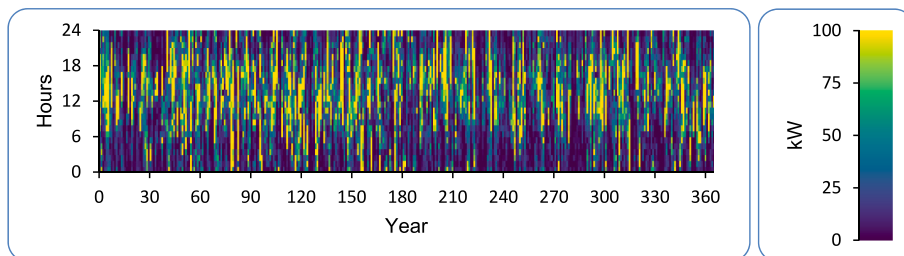


Fig. 15. Annual power output from wind turbine over 24 h of the day of the optimum hybrid energy system.

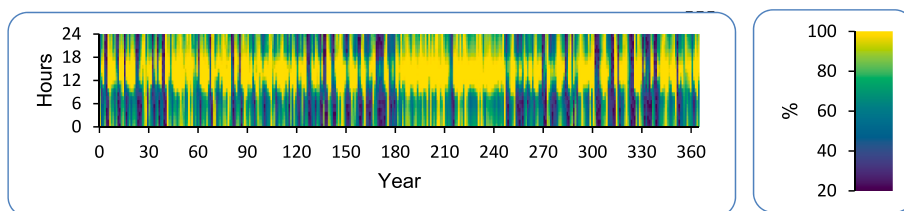


Fig. 16. Annual state of charge (in percent) of lithium-ion batteries over 24 h daily of the optimum hybrid power system.

MOHAMED MAHMOUD SAMY: substantial contribution to analysis and interpretation of data; MOHAMED MAHMOUD SAMY: drafting the article; SHIMAA BARAKAT: critically revising the article for important intellectual content; AMRO AMEEN: final approval of the version to be published; SHIMAA BARAKA: final approval of the version to be published; MOHAMED MAHMOUD SAMY: final approval of the version to be published; ELSAYED TAG-ELDIN: final approval of the version to be published.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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